

AI-Powered Bid & Proposal Response Engine

Industry Vertical: Procurement, Sourcing & Contract Management

Difficulty Level: Advanced

Team Size: up-to 3 Members

Background & Problem Context

Organizations in the public and private sector issue thousands of Requests for Proposals (RFPs), Requests for Quotations (RFQs), and Tenders every year. Bid and proposal managers spend 60–80% of their time reading lengthy tender documents, extracting compliance requirements, cross-referencing internal capability libraries, writing narrative sections, and formatting final submissions — a highly repetitive, error-prone, and time-intensive process.

A typical enterprise bid team handles 40–120 bids per year. Each bid document can range from 50 to 500+ pages. Missing a single mandatory compliance clause or submitting a poorly structured technical narrative can result in disqualification — directly impacting revenue. There is a clear need for an intelligent assistant that can parse bid documents, identify requirements, match them to organizational capabilities, and draft high-quality, compliant proposal responses automatically.

Core Challenge

Build an AI-powered Bid Response Engine that can:

- Ingest an RFP/RFQ/Tender document (PDF or DOCX) and automatically extract: mandatory requirements, evaluation criteria, submission deadlines, and question/answer sections.
- Match extracted requirements against a pre-loaded Company Capability Library (past project summaries, certifications, CVs, case studies).
- Auto-draft a structured proposal response document with relevant content mapped to each question/section.
- Flag compliance gaps where the organization lacks evidence or capability, allowing the bid manager to prioritize effort.
- Score the bid opportunity using win-probability heuristics (budget alignment, competitor presence, past win rate in similar domains).

Expected Outcomes / Deliverables

- A working prototype / POC that accepts a sample RFP document and generates a structured draft response.
- Separate workspace for each RFP/RFQ/Tender
- A compliance checklist auto-generated from the bid document with pass/fail mapping against the capability library
- A win-probability dashboard showing bid scoring across key criteria.
- Make GO/NO-GO decision

- Demonstrated reduction in manual bid preparation effort by at least 50% (measured against a baseline manual exercise).
- An intuitive UI allowing bid managers to review, edit, and approve AI-generated content before final export.

AI Components Required

- Large Language Model (LLM) for document parsing, requirement extraction, and narrative generation.
- Retrieval-Augmented Generation (RAG) to query the Capability Library for matching evidence.
- Named Entity Recognition (NER) to extract deadlines, budget figures, evaluation weights, and compliance clauses.
- Scoring/ranking model for win probability assessment.

Sample Dataset Needed

The following sample data should be provided to teams for prototyping:

Dataset	Description	Source Document	Size
Sample RFP Documents	3 anonymized government/enterprise RFPs (IT services, construction, logistics) – can be downloaded from Karandaz website	PDF/DOCX	15–80 pages each
Historical Bid Outcomes	Win/loss data for 120 past bids with evaluation scores	Sample Dataset Sheet	120 rows × 18 cols
Capability Library	50 past project summaries, included company certification, year completed, contract value, durations and client type	Sample Dataset Sheet	50 records
Evaluation Criteria Taxonomy	Structured list of 15+ common RFP evaluation criteria by sector	Sample Dataset Sheet	15 entries